习题解答

5-8：解： 缓冲对：HCN/CN^- 则有 $K_b^\ominus(CN^-) = K_w^\ominus / K_a^\ominus(HCN)$

$$K_b^\ominus(CN^-) = 1.0\times10^{-14} / 4.93\times10^{-10} = 2.03\times10^{-5}$$

	CN^-	+ H_2O =	HCN	+ OH^-
c_0	0.10		0	0
c_{eq}	$0.10 - y$		y	y

$c_0/K_b^\ominus(CN^-) > 500$，可近似计算。

$$K_b^\ominus = \frac{y \cdot y}{0.10 - y} \approx \frac{y^2}{0.10}$$

$$c(OH^-) = y = [0.10\times K_b^\ominus(CN^-)]^{1/2} = 1.42\times10^{-3}$$

$$pOH = -\lg 1.42\times10^{-3} = 2.85$$

$$pH = 14.00 - 2.85 = 11.15$$

习题解答

5-9：解：　$pH = -\lg c(H^+) = 4.5$　则有　$c(H^+) = 3.16\times10^{-5}$ mol/L

	HA	+	H_2O	=	A^-	+	H_3O^+
c_0	0.010				0		0
c_{eq}	$0.010-3.16\times10^{-5}$				3.16×10^{-5}		3.16×10^{-5}

可近似计算：

$$K_a^\Theta = \frac{(3.16\times10^{-5})\cdot(3.16\times10^{-5})}{0.010-3.16\times10^{-5}} \approx \frac{(3.16\times10^{-5})^2}{0.010} = 1.0\times10^{-7}$$

$$\alpha = 3.16\times10^{-5}/0.010 = 0.316\%$$

5-11：解：　$\alpha = c(H^+)/c_0 = 0.042$；　$c(H^+) = 4.2\times10^{-4}$ mol/L

$$K_a^\Theta = [c(H^+)]^2/[c_0 - c(H^+)] = (4.2\times10^{-4})^2/0.01 = 1.76\times10^{-5}$$

习题解答

5-14：解：　设加入NaAc固体 x g

则：$n(\mathrm{HAc}) = 0.50\times300/1000$ mol;

$n(\mathrm{Ac^-}) = x/82$ mol

$$\mathrm{pH} = \mathrm{p}K_\mathrm{a}^\ominus(\mathrm{HAc}) - \lg\frac{c(\mathrm{HAc})}{c(\mathrm{Ac^-})} = \mathrm{p}K_\mathrm{a}^\ominus(\mathrm{HAc}) - \lg\frac{n(\mathrm{HAc})}{n(\mathrm{Ac^-})}$$

$$4.50 = 4.74 - \lg\frac{0.50\times\dfrac{300}{1000}}{\dfrac{x}{82}}$$

$x = 7.1$　　g

习题解答

5-15：解：

(1) HCl过量

$$c(\mathrm{H^+}) = [0.10\times 3V - 0.10\times 2V]/5V = 0.020 \quad \mathrm{mol\cdot L^{-1}}$$

$$\mathrm{pH} = 1.7$$

(2)

$$\mathrm{pH} = \mathrm{p}K_a^{\ominus}(\mathrm{HAc}) - \lg\frac{c(\mathrm{HAc})}{c(\mathrm{Ac^-})}$$

$$= \mathrm{p}K_a^{\ominus}(\mathrm{HAc}) - \lg\frac{0.10\times 3V - 0.10\times 2V}{0.10\times 2V}$$

$$= 4.74 - \lg 0.5$$

$$= 5.04$$

习题解答

$$(3)\ \mathrm{pH} = \mathrm{p}K_a^{\ominus}(\mathrm{NH_4^+}) - \lg\frac{c(\mathrm{NH_4^+})}{c(\mathrm{NH_3})}$$

$$= \mathrm{p}K_a^{\ominus}(\mathrm{NH_4^+}) - \lg\frac{0.10\times 2V}{0.10\times 3V - 0.10\times 2V}$$

$$= 9.25 - \lg 2$$

$$= 8.95$$

$$(4)\ \mathrm{pH} = \mathrm{p}K_a^{\ominus}(\mathrm{HPO_4^{2-}}) - \lg\frac{c(\mathrm{HPO_4^{2-}})}{c(\mathrm{PO_4^{3-}})}$$

$$= \mathrm{p}K_a^{\ominus}(\mathrm{HPO_4^{2-}}) - \lg\frac{0.10\times 2V - 0.10\times 1V}{0.10\times 1V}$$

$$= 12.66 - \lg 1$$

$$= 12.66$$

习题解答

5-17：解：HA过量

$$\mathrm{pH} = \mathrm{p}K_{\mathrm{a}}^{\ominus}(\mathrm{HA}) - \lg\frac{c(\mathrm{HA})}{c(\mathrm{A}^-)}$$

$$5.25 = \mathrm{p}K_{\mathrm{a}}^{\ominus}(\mathrm{HA}) - \lg\frac{\dfrac{50\times0.10-20\times0.10}{100}}{\dfrac{20\times0.10}{100}}$$

$$\mathrm{p}K_{\mathrm{a}}^{\ominus}(\mathrm{HA}) = 5.25 + \lg\frac{3}{2}$$

$$K_{\mathrm{a}}^{\ominus}(\mathrm{HA}) = 3.72\times10^{-6}$$

习题解答

5-18：解：混合后生成0.010 mol·L^{-1}的$C_6H_5NH_3^+$(弱酸)

$$K_a^\Theta(C_6H_5NH_3^+) = K_w^\Theta / K_b^\Theta(C_6H_5NH_2)$$

$$= 1.0\times10^{-14}/4.6\times10^{-10}$$

$$= 2.2\times10^{-5}$$

$c_0/K_a^\Theta > 500$，可以用近似式计算：

$$c(H^+) = (0.010\times K_a^\Theta)^{1/2} = (0.010\times2.2\times10^{-5})^{1/2}$$

$$= 4.7\times10^{-4}$$

$$pH = -\lg 4.7\times10^{-4} = 3.3$$